



Neural Spike Train Classification

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Methods and Metrics



- **Classifiers used:** Logistic Regression, Random Forest
- **Data:** Spike trains (800 samples, 1000 neurons, 501 timesteps)
- **Preprocessing:** Flattening spike train data
- **Evaluation metrics:** Accuracy, F1 Score, Confusion Matrix





Results



- **Logistic Regression:** ~73.8% accuracy, F1 score ~0.71
- **Random Forest:** ~92.0% accuracy, F1 score ~0.92
- **Confusion matrices** show **strong performance of Random Forest**





Interpretation



- Random Forest significantly outperforms Logistic Regression
- Spike train data appears to contain non-linear patterns better captured by ensemble methods
- Suggests future work could include tuning RF or exploring deep learning

